Matrices, Vectors & Complex Numbers

Matrix arithmetic

1 Matrix A has rows $(2, 3)$ and $(-1, 4)$; matrix B has rows $(5, -2)$ and $(1, 3)$. Find the rows of $A + B$.

2 Find $3A$ for matrix A with rows $(1, -2)$ and $(4, 0)$.

Matrix multiplication

3 Matrix A has rows $(1, 2)$ and $(3, 4)$; matrix B has rows $(5, 6)$ and $(7, 8)$. Find AB.

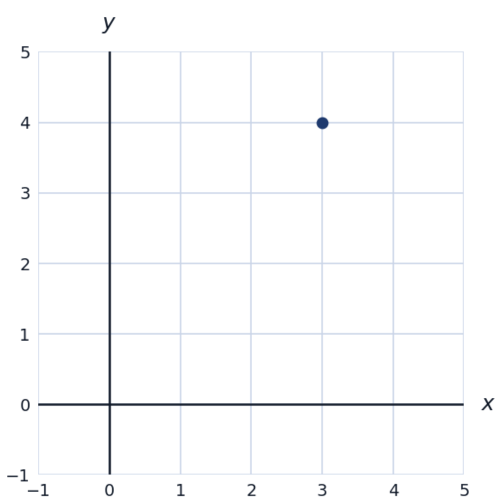
Determinants and inverses of 2×2 matrices

4 Find the determinant of the matrix with rows $(4, 2)$ and $(3, 5)$.

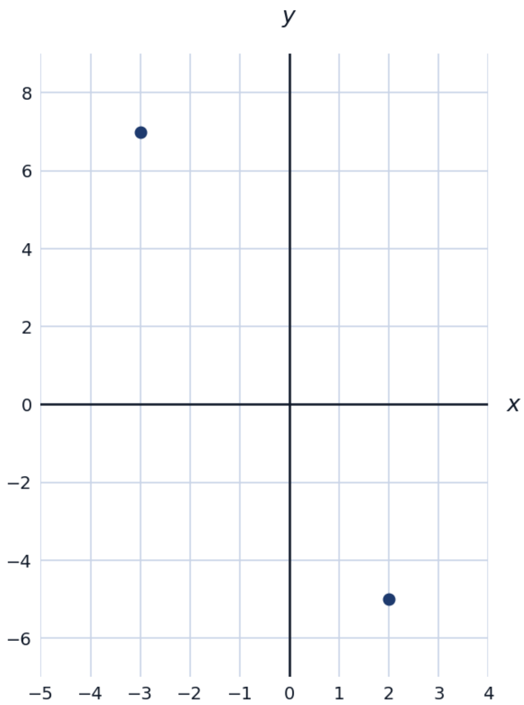
5 Find the inverse of the matrix M with rows $(2, 1)$ and $(5, 3)$.

Vectors: magnitude and direction

6 Find the magnitude of the vector $\mathbf{v} = (3, 4)$.

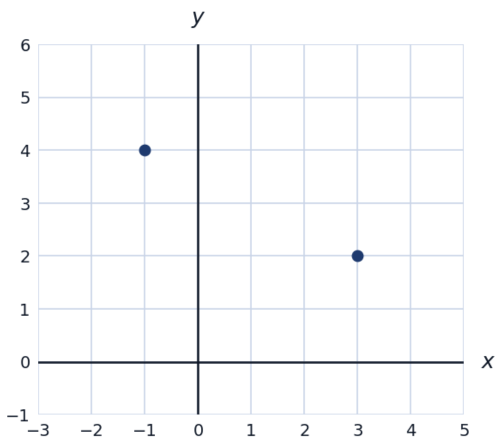


7 Given $\mathbf{a} = (2, -5)$ and $\mathbf{b} = (-3, 7)$, find $\mathbf{a} + \mathbf{b}$ and $2\mathbf{a} - \mathbf{b}$.



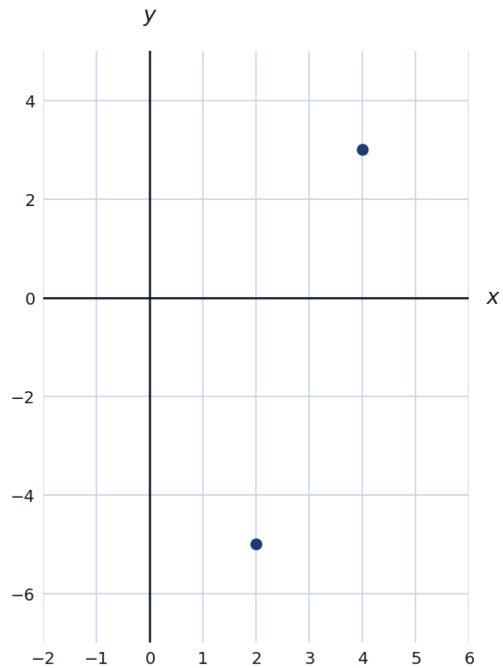
The dot product

8 Find $\mathbf{a} \cdot \mathbf{b}$ for $\mathbf{a} = (3, 2)$ and $\mathbf{b} = (-1, 4)$, and determine whether the vectors are perpendicular.

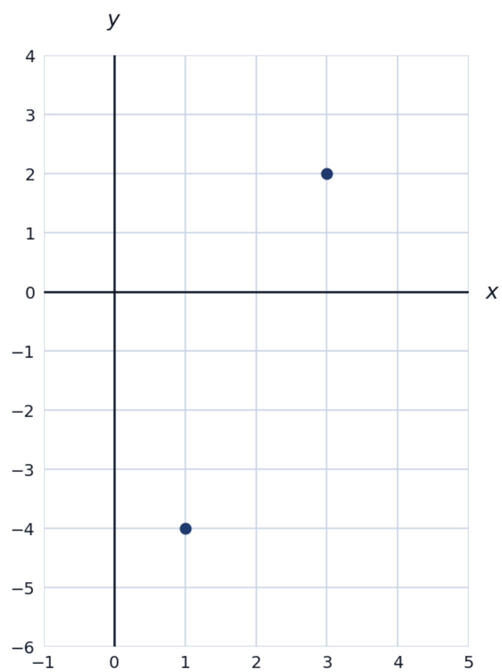


Complex number arithmetic

9 Simplify $(4+3i)+(2-5i)$.

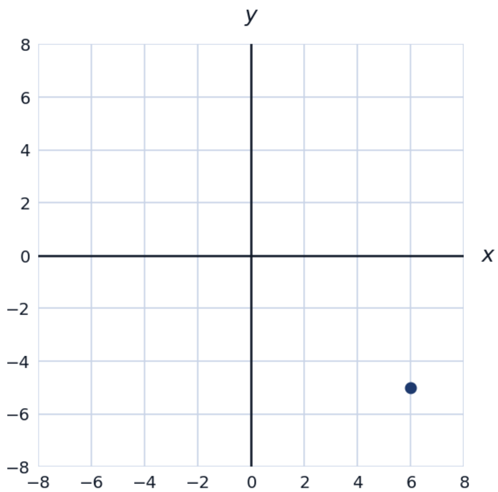


10 Simplify $(3+2i)(1-4i)$.

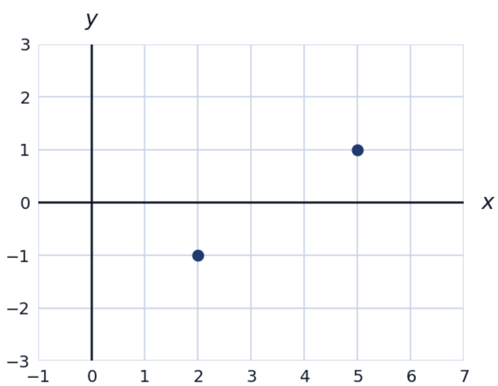


Complex conjugates and division

11 Find the complex conjugate of $z = 6 - 5i$, and compute $z\bar{z}$.

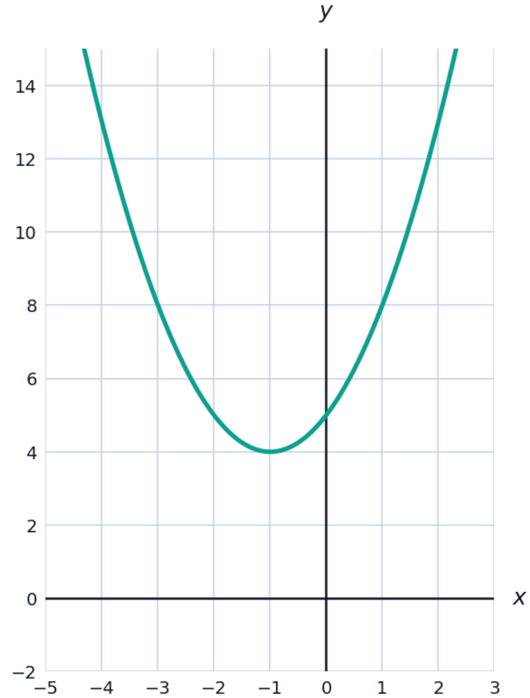


12 Simplify $\dfrac{5+i}{2-i}$ by multiplying by the conjugate of the denominator.



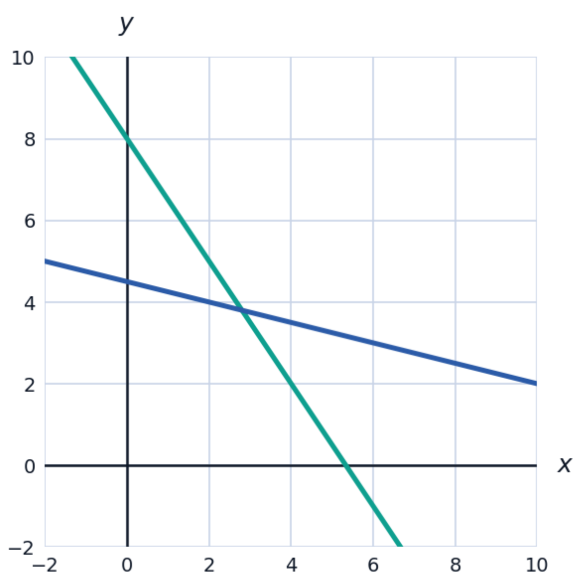
Solving quadratics with complex solutions

13 Solve $x^2 + 2x + 5 = 0$, giving your answer in the form $a + bi$.



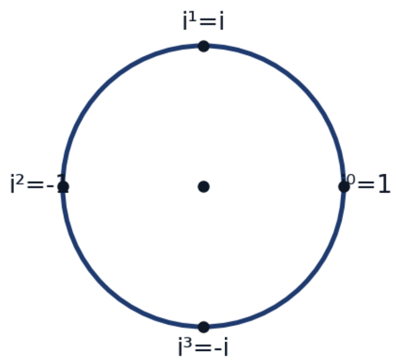
Combining matrices and systems of equations

14 This question has two parts. The system $3x + 2y = 16$ and $x + 4y = 18$ can be written as a matrix equation $M\mathbf{v} = \mathbf{c}$, where M has rows $(3, 2)$ and $(1, 4)$. (a) Find $\det(M)$ and M^{-1}. (b) Use M^{-1} to solve the system for x and y.



Powers of i and simplifying complex expressions

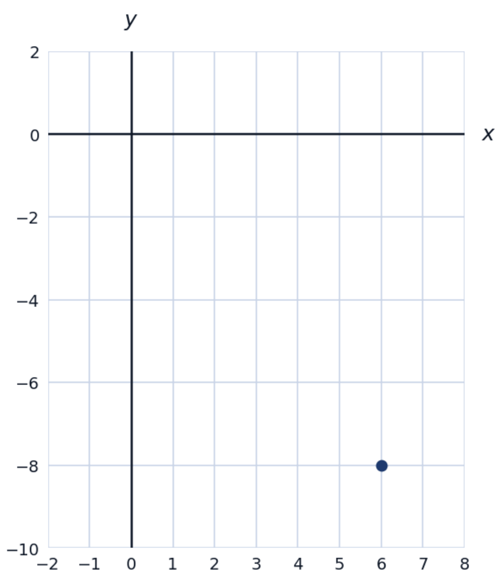
15 Simplify i^{17}, using the fact that powers of i cycle with period 4.



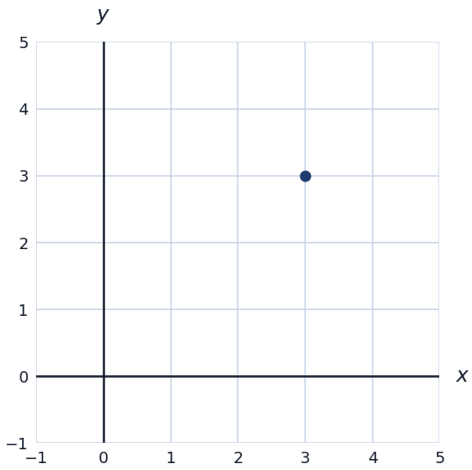
16 Simplify $(2i)^3$.

Vector direction and unit vectors

17 Find a unit vector in the same direction as $\mathbf{v} = (6, -8)$.



18 Find the angle a vector $\mathbf{v} = (3, 3)$ makes with the positive x-axis.

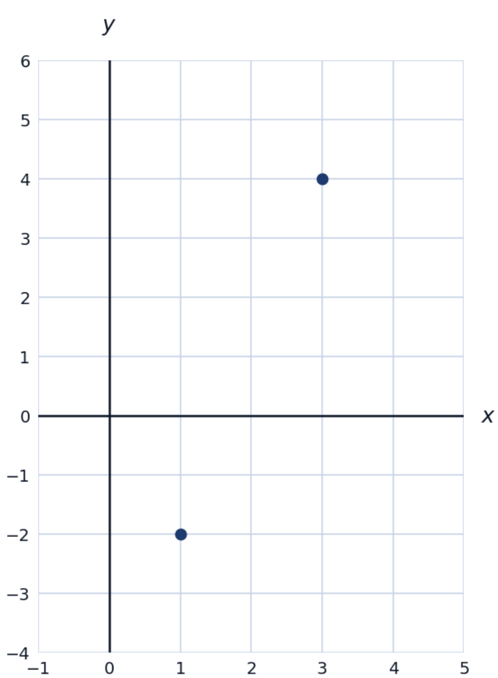


Identity matrix and matrix equations

19 Explain why $MI = M$ for any matrix M and identity matrix I of matching size, and verify this for M with rows $(2, 3)$ and $(1, 4)$ multiplied by the 2×2 identity matrix (rows $(1, 0)$ and $(0, 1)$).

A multi-part complex number problem

20 This question has two parts. Given $z_1 = 3 + 4i$ and $z_2 = 1 - 2i$. (a) Find $|z_1|$ (the modulus of z_1). (b) Find $z_1 \cdot z_2$ and its modulus, then verify that $|z_1 \cdot z_2| = |z_1| \cdot |z_2|$.



Scalar multiplication of matrices and vectors combined

21 Given vectors $\mathbf{u} = (4, -3)$ and $\mathbf{v} = (-2, 5)$, find $3\mathbf{u} - 2\mathbf{v}$.